PROGRAMMING FUNDAMENTAL

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ASSIGNMENT 01

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<u>QUESTION 01:</u> Define and explain programming paradigm with the help of examples.

ANSWER:

PROGRAMMING PARADIGMS:

Programming paradigms refers to the style, way or approach to organize a program and solve a problem using different programming languages.Each paradigm consist of certain structures and features about how common programming problems should be tackled

TYPES OF PROGRAMMING PARADIGMS:

1.Imperative programming paradigm:

In imperative programming paradigm, programmer instructs the computer how to change its state. It determines its path by changing its state step by step according to the assignment statements.The paradigm consist of several statements and after execution of all result is stored.
Examples include : C, C++, Java, Python etc.
Imperative paradigm is further divided into three categories:

- <u>Procedural programming</u>

  Procedural programming paradigm is based on the use of procedure calls.Procedures are functions, routines or subroutines that specify the computaional steps that need to be performed.

- <u>Object oriented programming</u>

  Object oriented programming couples data and methods together into objects. OOP deals with entities. The smallest and basic entity is called object. Objects respond to messages by performing operations generally called methods.

- <u>Parallel programming</u>

  Parallel programming is the processing of program instructions by dividing them among multiple processors that run simultaneously.

Example of imperative paradigm:

```
int a = 5;        // Step 1: assign value
int b = 10;       // Step 2: assign value
int c = 15;       // Step 3: assign value
int sum = a + b + c;  // Step 4: compute sum
System.out.println(sum); // Step 5: output result
```

2.Declarative programming paradigm:

In declarative programming paradigm programmer describes the desired results and outcomes rather than specifying the step by step instructions. It just tells what to do rather than how to do. For e.g: SQL, Prolog etc
Declarative paradigm is divided into these categories:

- Functional Paradigm

  In Functional Paradigm programs are constructed by composing functions. All code is within the function.

- Logic Paradigm

  Logic paradigm composes program using logic circuits and to solve problems via logical deductions.

- Database Programming Approach

  In Database Programming users declare the data they want not how to get that.

Example of Declarative Paradigm:

```
int max = Math.max(10,20);
  System.out.print("Maximum num is: " + max);
```

QUESTION 02

a. Write java statements that can produce Syntax Errors.Give three different examples and write the names of errors

SYNTAX ERRORS:

- Missing a Semi-colon:

  ```
  System.out.print("Hello World")
  ```

- Mistyped Keyword:

  ```
  public Static Void main(String[ ] args) {
  ```

- Mismatched Brackets:

  ```
  System.out.print("Hello World"}
  ```

b. Write java statements that can produce Logical Errors. Give three different examples and write the names of errors.

LOGICAL ERRORS:

- Incorrect Formula:

  Int area = length + width;

  Using this formula will produce incorrect results as correct formula is length*width.

- Operator precedence order;

  Int avg = a+b+c/2

  Using this formula will lead to division only of c not the sum of a b and c.

- Wrong Operator:

  int sum = a-b;

  The variable is declared as sum so using “-” operator will perform wrong calculations.

c. Write java statements that can produce Runtime Errors.Give three examples and briefly explain the reason.

RUNTIME ERRORS:

- Division by zero:

  double div = a/0;

  Division by zero is undefined in mathematics so it will lead to runtime errors

- Wrong input entry:

  System.out.print(“Enter a number: “);

  int num = sc.nextInt();

  Entering a String value in this datatype will lead to runtime errors

- Input/Output Errors:

  Happen when user input or file operations go wrong.
  Example: Asking the user for a number but they type text instead.

d. The following programs have errors. Write clearly the types of errors and its correction in tabular form. Also show the output.

PROGRAM 01

ERRORS	ERROR TYPE	PROBLEM	CORRECTION
1. int number; number = “ten”;	Syntax Error	String is declared in integer datatype.	int number; number = 10;
2. float = 3,1416;	Syntax Error	Wrong float declaration	float = 3.1416f;
3. double result == 0;	Syntax Error	== is comparison operator and here assignment(=) is required	double result =0
4. if (number = 5)	Syntax Error	== comparison operator should be used	If (number == 0)
5. (“Result is: “ + result;	Syntax Error	Closing bracket is missing	(“Result is: “+ result);

```
PS C:\Users\Zoha Khalid\desktop> javac ErrorDemo.java
PS C:\Users\Zoha Khalid\desktop> java ErrorDemo
Result is: 0.0
```

PROGRAM 02

ERRORS	**ERROR TYPE**	**PROBLEM**	**CORRECTION**
count = 1;	Syntax Error	Variable is not declared	int count =1;
sum = count +PRIME;	Syntax Error	Sum and PRIME are not declared as variable	int PRIME = 7; int sum = count + PRIME;
x := 25.67;	Syntax Error	**:=** is wrong assignment operator	x = 25.67;
newNum = count*ONE +2;	Syntax Error	newNum and ONE are not declared as variable	int ONE = 1; int newNum = count*ONE+2;
x = x + sum*COUNT;	Syntax Error	Java is case sensitive so code will not run with COUNT	x = x+sum*count;
“,PRIME = “ + Prime);	Syntax Error	Variable is declared as PRIME	**“,** PRIME = “ + PRIME);

OUTPUT:

```
PS C:\Users\Zoha Khalid\desktop> javac Test.java
PS C:\Users\Zoha Khalid\desktop> java Test
 count = 1, sum = 9, PRIME = 7
```

<u>QUESTION 03</u> :Identify the mentioned things in the given program

CATEGORY	LINES FROM CODE

1. Comments	MULTI LINE:
	/*This program will calculate product of three numbers */
	
	SINGLE LINE:
	// first number
	// second number
	// third number
	//product of numbers

2. Special symbols	; = statement terminator { } = define a block * = operator of multiplication
3. Reserve words	public class static
4. Identifiers	Defined by user: Product num1 result Predefined: System out print
5. Standard Input Stream Object	System.in
6. Standard Output Stream Object	System.out

QUESTION 04 :

a. Write java statements that accomplish the following.

1. Declare a String variable named name and assign it the value “Java”.

STATEMENT:

```
String name = "Java";
```

2. Declare a boolean variable named isPassed and set it to true.

STATEMENT:

```
boolean isPassed = true;
```

3. Multiply two int variables a and b, and store the result in an int variable product.

STATEMENT:

```
int a = 2;
int b = 3;
int product = a*b;
```

4. Declare a float variable temperature and initialize it with the value 36.6.

STATEMENT:

```
float temperature = 36.6f
```

5. Assign the remainder when dividing num1 by num2 to an int variable remainder.

STATEMENT:

```
int num1 = 4;
int num2 = 2;
int remainder = num1%num2;
```

6.Suppose p and q are double variables.Output the contents of p, q and the expression(p*q)/2.

STATEMENT:

```
double p = 6.6;
double q = 3.3;
double result = (p*q)/2;
```

System.out.print(“p: ” + p + “q: “ + “(p*q)/2 : “ + result;

7.Declare a char variable symbol and assign it the value ‘#’.

STATEMENT:

char variable =’ #’;

8. Declare three int variables to store the marks of three subjects.

int chem;

int phy;

int bio;

9. Copy the value of an int variable score into a double variable resultScore.

int score = 34;

double resultScore = (double) score;

10. Suppose radius is a double.Write a statement to compute the area of a circle and store it in a double variable area.

STATEMENT:

double radius = 6.28;

final double PI = 3.1416;

double area = PI * radius * radius;

b. Suppose a, b, c, d, are int variables and a = 5, b= 6, c=7, d=2. What value is assigned to each variable after each statement executees?If a variable is undefined at a particular statement, report UND.

STATEMENTS	a	b	c	d
a = (++b) * 2 + (c--);	21	7	6	2
c = (a++) - (--d) + b;	22	7	27	1
b = (d--) + (c++) * ++a;	23	622	28	0
d = (--a) + (b++) - (c--);	22	623	27	616

c. Suppose a, b, and sum are int variables and c is a double variable. What value is assigned to each variable after each statement executes? Suppose a = 3, b = 5 , and c = 14.1

STATEMENTS	a	b	c	sum
sum = a + b + (int) c;	3	5	14.1	22
c /= a;	3	5	4.7	22
b += (int) c - a;	3	6	4.7	22
a *= 2 * b + (int) c;	48	6	4.7	22

QUESTION 05 : Ali lives in a village where he uses both solar energy and electricity from the company. During the day, he consumes electricity generated from his solar system, which costs him only Rs. 7 per unit. After 5 PM, he uses electricity from the company, which charges Rs. 60 per unit. Ali wants to calculate how much money he is saving by using solar energy. Create a program that asks for the number of units consumed from the solar system and the number of units consumed from the electricity company. The program should then calculate the total bill if all units were taken from the company, the actual bill using both solar and company electricity, and the total savings Ali makes by using solar power.

PROGRAM:

```
import java.util.Scanner;

public class Bill {

    public static void main(String [] args) {

    Scanner console = new Scanner(System.in);

     int solarSystem = 7;

     int company = 60;


    System.out.print("Enter no. of units consumed from solar system: ");

     int solarUnit = console.nextInt();
```

```
    System.out.print("Enter no. of units consumed from comapny: ");
    int companyUnit = console.nextInt();

    int totalUnits = solarUnit + companyUnit;
    int solarBill = solarUnit * solarSystem;
    int companyBill = companyUnit * company;

    int actualBill = solarBill + companyBill;
    int totalCompany = company * totalUnits;

    int savings = totalCompany - actualBill;

    System.out.println("The actual bill is " + actualBill + " Rs ");
     System.out.println("The total bill if all units were from company is " + totalCompany +
" Rs ");
    System.out.println("The savings are " + savings + " Rs ");
  }
 }
```

```
Enter no. of units consumed from solar system: 14
Enter no. of units consumed from comapny: 16
The actual bill is 1058 Rs
The total bill if all units were from company is 1800 Rs
The savings are 742 Rs
```

QUESTION 06 : Sara visits her doctor for a regular health check-up, and the doctor asks her to calculate her Body Mass Index (BMI) to monitor her fitness. Sara's weight is 62 kilograms, and her height is1.68 meters. The formula for BMI is the weight divided by the square of the height. Create a program that calculates Sara's BMI and displays the result.

PROGRAM:

```
public class BMI {
  public static void main(String[] args) {
  double weight = 62;
  double height = 1.68;
  double BMI = weight/(height*height);
  System.out.printf("Sara's Body Mass Index is: %.2f%n  " ,  BMI);
 }
}
```

```
PS C:\Users\Zoha Khalid\desktop> javac BMI.java
PS C:\Users\Zoha Khalid\desktop> java BMI
Sara's Body Mass Index is: 21.97
```

QUESTION 07 : Ahmed works in a private company and earns Rs. 50,000 each month. His monthly expenses, however, are Rs. 37,500. Ahmed plans to purchase a new laptop that costs Rs. 100,000. He wants to calculate how long it will take to save enough money. Create a program that calculates Ahmed's monthly savings and determines how many months are required for him to save enough money to buy the laptop.

PROGRAM:

```
public class Laptop {
   public static void main(String[] args) {
   int earning = 50000;
   int expenses = 37500;
   int savings = earning - expenses;
   int cost = 100000;
    int months = cost/savings;
   System.out.print("Months required for him to save enough money is " + months);
```

```
 }
}
```

```
PS C:\Users\Zoha Khalid\desktop> javac Laptop.java
PS C:\Users\Zoha Khalid\desktop> java Laptop
Months required for him to save enough money is 8
```

QUESTION 08 : The weather station in Lahore has reported today's temperature as 32 degrees Celsius. For preparing an international weather report, the same temperature also needs to be expressed in Fahrenheit and Kelvin. The formulas to use are Fahrenheit = (9/5 × Celsius) + 32 and Kelvin = Celsius + 273.15. Create a program that converts 32 degrees Celsius into Fahrenheit and Kelvin and displays the converted values

PROGRAM:

```
public class Temperature {
   public static void main(String[] args) {
   int Celsius = 32;
   double Fahrenheit = (Celsius*9/5)+32;
   double Kelvin = Celsius + 273.15;
   System.out.println("32 Celsius in Fahrenheit is " + Fahrenheit);
   System.out.println("32 Celsius in Kelvin is " + Kelvin);
  }
 }
```

```
PS C:\Users\Zoha Khalid\desktop> java Temperature
32 Celsius in Fahrenheit is 89.0
32 Celsius in Kelvin is 305.15
```

QUESTION 09 : Hassan has purchased a car by taking a loan of Rs. 1,200,000 from the bank. The loan is to be repaid in 5 years with an annual interest rate of 12%. Hassan wants to know his monthly installment amount so he can plan his expenses.

The formula for calculating the monthly installment is Payment = (P × r) / (1 – (1 + r)^(-n)), where P is the loan amount, r is the monthly interest rate (annual rate divided by 12), and n is the total number of months. Create a program that calculates Hassan's monthly car loan installment

PROGRAM:

```
public class Loan {
   public static void main(String[] args) {    int n = 60;
   int P = 1200000;
   double r = 0.12/12;

   double monthly = (P*r)/(1-Math.pow(1+r,-n));
   System.out.printf("Hassan's monthly car loan installment is Rs. %.2f%n", monthly);
 }
}
```

```
PS C:\Users\Zoha Khalid\desktop> java Loan
Hassan's monthly car loan installment is Rs. 26693.34
```

QUESTION 10 : Research several car-pooling websites. Create an application that calculates your daily driving cost, so that you can estimate how much money could be saved by carpooling, which also has other advantages such as reducing carbon emissions and reducing traffic congestion. The application should input the following information and display the user's cost per day of driving to work:

a) Total miles driven per day.

b) Cost per gallon of gasoline.

c) Average miles per gallon.

d) Parking fees per day.

e) Tolls per day.

PROGRAM:

```
import java.util.Scanner;
  public class Driving {
```

```
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);

System.out.print("Enter total miles driven per day: ");
double totalMiles = sc.nextDouble();

System.out.print("Enter cost per gallon of gasoline: ");
double gallonCost = sc.nextDouble();

System.out.print("Enter average miles per galon: ");
double averageMilespergallon = sc.nextDouble();

System.out.print("Enter parking fees per day: ");
double parkingFees = sc.nextDouble();

System.out.print("Enter tolls per day: ");
double tolls = sc.nextDouble();

double fuelCost = (totalMiles/averageMilespergallon)*gallonCost;
double drivingCost = fuelCost + parkingFees + tolls;

    System.out.printf("The user’s cost per day of driving to work is %.2f%n " , drivingCost);
 }
}
```

```
Enter total miles driven per day: 12
Enter cost per gallon of gasoline: 500
Enter average miles per galon: 6
Enter parking fees per day: 200
Enter tolls per day: 100
The user?s cost per day of driving to work is 1300.00
```

QUESTION 11 : Write an application that inputs one number consisting of five digits from the user and show the number in reverse order. Note the reverse order must also be a number. E.g. 93324

PROGRAM:

```
import java.util.Scanner;
public class Num {
  public static void main(String[] args) {
  Scanner sc = new Scanner(System.in);

  System.out.print("Enter a five digit number: ");
  int num = sc.nextInt();
  int reverse;
  int digit1 = num/10000;
  int digit2 = (num/1000)%10;
  int digit3 = (num/100)%10;
  int digit4 = (num/10)%10;
  int digit5 = num%10;

  reverse = (digit5*10000)+(digit4*1000)+(digit3*100)+(digit2*10)+(digit1);
  System.out.println("Original number is: " + num);
  System.out.println("Reverse order is: " + reverse);
```

```
  }
}
```

```
PS C:\Users\Zoha Khalid\desktop> java Num
Enter a five digit number: 23456
Original number is: 23456
Reverse order is: 65432
```